

Ethan Wong

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TECHNICAL SKILLS

Design: Altium, KiCad, Quartus, VScode, FPGA, Solidworks, Fusion360, Excel

Programming: C/C++, Python, MATLAB, System Verilog, Verilog, RISC V Assembly

Hardware: Soldering (SMD/THT), Oscilloscope, STM32, CAN, SPI, I2C, STM32, Raspberry Pi, Arduino

EDUCATION

University of British Columbia

Bachelor of Applied Science, Electrical Engineering

Co-op: Available for 4, 8, or 12 Months

Expected Graduation: Apr. 2029

CGPA: 4.00/4.33

ENGINEERING STUDENT TEAMS

UBC Solar Car - Senior Power and Signals Engineer

September 2025 - Present

- Designed, simulated, and validated multi-layer PCBs from schematic to production, implementing grounding strategies, decoupling techniques, controlled impedance routing, and validated CAN signal timing and I2C bus hardware to ensure signal integrity in mixed-signal electrical systems
- Conducted rigorous component selection and system integration analysis, evaluating microcontroller architectures, device integration, and electrical characteristics to meet hardware and firmware requirements
- Performed board bring-up and functional validation of assembled PCBs, hand-soldering SMD and THT components, systematically diagnosing hardware faults using oscilloscopes and multimeters, iterating on design revisions to resolve signal integrity and communication issues
- Authored comprehensive electrical system documentation including circuit design specifications, component selection rationale, validation results, and troubleshooting procedures for multi-disciplinary integration
- Led structured design reviews with the Firmware team, aligning hardware-software interfaces and resolving integration issues through iterative testing and validation
- Mentored new team members in PCB design principles, schematic capture, component selection, and board layout practices in compliance with UBC Solar standards

PROJECTS

Motor Controller Interface PCB | *Altium Designer, STM32, CAN, I2C, SPI*

November 2025 - April 2026

- Designed a 4-layer PCB in Altium Designer, capturing schematics and laying out the board to interface the M2096 Mitsuba Motor Controller across 16 signals including speed, direction, and torque, using I2C-controlled DACs and a CAN transceiver on the STM32F103RCT6
- Ensured analog signal integrity through stitching vias and a continuous ground plane, and applied differential routing with isolation techniques for CAN traces
- Applied differential signal routing for CAN traces, including effective isolation techniques

Telemetry Board PCB | *Altium Designer, STM32, XBEE Radio, GPS, IMU*

February 2026 - April 2026

- Interfaced the XBEE Radio module or NETGEAR cellular hotspot through UART to transmit CAN Bus data over the air to InfluxDB for display on Grafana
- Communicated with the NEO-M9V-20B GPS module using I2C, incorporating multiple GPIOs including a toggleable external wheel tick from either the STM32 microcontroller or the Motor Controller.
- Received data from the BNO086 IMU through I2C communication and accompanying initialization pins for the IMU including BOOT, reset, and initialization
- Provided a connection to an external Memorator to log CAN data across the car and a USB-PD connection to supply sufficient power to a Raspberry PI for Cellular communication

Remote Control and Magnetic Field Following Robot | *Embedded C, STM32, EFM8, Altium*

March 2026

- Developed embedded firmware for servo motor control using 2kHz PWM generation through STM32CubeMX and dynamically adjusting duty cycle by writing to CCR registers
- Designed and integrated peripheral subsystems on a custom remote control PCB, including 555 timer circuit IR transmission, dual LCD interface, UART ports, and user I/O
- Engineered a multi-system robot PCB incorporating IR and NRF24L01 2.4GHz radio communications, a discrete MOSFET H-Bridge motor driver with optocoupler isolation, analog signal readings using op-amp inductor circuits, and dual-input galvanic isolation for microcontroller and motor power